

JK 12-16 Python First Programmer
Lesson 08 : Random Module Projects

Duration : 90 mins

Description

Students create projects incorporating dice capability using random module, which is generating a random number between 1 to 6. This lesson will put into practice all the knowledge that students have gained during lesson 1 to lesson 7. There are several projects for this lesson and each project complexity can be adjusted to students' understanding so far.

Objectives

1. Re-Introduce students to the concept of random numbers and how to implement it in a projects.
2. Create a few simple projects to re-introduce the concept of random before we jump into the main project.
3. Analyze the codes and figure out the outcome of the codes.
4. Create projects combining all concepts learned during the lesson.

Tools

1. Pycharm

Class Structure

Duration	Topic
5 mins	Preparation
5 mins	Comprehension Check
5 mins	Randomness
5 mins	Head or Tail
15 mins	Dice Game
15 mins	Advanced Dice Game
20 mins	Rock Paper Scissor (No Function)
15 mins	Rock Paper Scissor (With Function)
5 mins	Wrap Up

Planned Activities	Resources
Preparation (5 mins)	
- Teacher greets students and ask how's their conditions. - Instruct students to open their pycharm. - If possible, teacher can incorporate a mini game during this preparation time to engage with the students.	
Comprehension Check (5 mins)	
- Teacher ask students if they are still remember about the materials from the previous lesson. - Teacher check on students' comprehension regarding the previous lesson by asking relatable questions. - Teacher explain and demonstrate in brief or summarize about what they have learned from the previous lesson.	
Randomness (5 mins)	
Discuss and Explain – Randomness and The Importance of Randomness - We start by letting the students know that for today's topic, we will explore further the capabilities of randomization. - We ask students if they still remember what we could use a randomized number for and show students some examples on how random might be used in programs. For example, if we are doing guess the number game, it will be really boring if we know what the answer is every time	

we play it. Therefore, inside the program, there must be a randomization component to randomize the game and make it more interesting.

Head or Tail (5 mins)

Explain and Code – Head or Tail

- We ask students if they still remember how to add randomness to their project. If students forgot how to add it, we can explain on the big screen that they can use the `random.randint(num1, num2)` function to randomize the numbers. We remind students that if they want to add the random function, we have to remember to import the random function into our python program.
- We tell students to create a simple project using the `random.randint` function. To give them an example, we can show the students the coin flip project. We explain to them that by randomizing between 1 & 2, we can then use conditional statements to show the results. In this case, head or tail.

File/Code:

[1.HeadOrTail.py](#)

Dice Game (15 mins)

Discuss and Explain – Rolling the Dice

- We explain to students that they will create a game that will involve rolling a dice as it is one of the most common ways to add some sort of randomness into our game.
- We show students the dice game project example and ask students what they think would happen in the project.
- We explain that in the project, the program will inform the player to press enter to roll the dice until the player inputs a “no” which will end the project execution.

File/Code:

[2.Dice.py](#)

Code and Explain – Roll the Dice Project

- We ask students if they have any guesses on how to create the project. We listen to their ideas and tell them to try creating it by themselves first.
- After we give some time to create the dice project, we ask them what functions they used and if they encountered some errors during the process.
- We code it together with the students' help on the big screen.

Advanced Dice Game (15 mins)

Explain and Show – Advanced Dice Game

- We tell students that for the next step, we will create a more complex dice game by using the basic concepts that they have previously learned.
- We show students the example and ask them what happens in the project. We analyze it together with the students.
- If there are some parts that are still not mentioned by the students, we should try and re-explain the process again. In general, there are **4 main points that happens in the project**:
 - **First**, in the beginning, **the program will let the player know** that the game will be about a simple dice **game** that will test our luck by seeing how many dice rolls it takes in order for the player to reach the finish line.
 - **Second**, the program will **continuously prompt a command for the player to roll the dice** (by pressing enter) until the player reaches the finish line that we have set.
 - **Third**, based on the score, if the player's score is higher than the **high score** that has been set, the player's score will then replace it and will be displayed on the screen.
 - **Fourth**, The program will **prompt** the user if they would like to **restart or not**. If the player says yes, the program will restart the game.

File/Code:

[3.Dice2.py](#)

Discuss and Code – Advanced Dice Game Project

- Step by step, we then explain our project to the students and ask them to add components that might be missing / if they are still confused, we can try to explain it slowly as the students try to follow.
- We re-explain the 4 main components to students and explain the flow of the program. Such as how we start by declaring all of the necessary variables, then printing out the texts, and then creating loops to prompt the players.

Rock Paper Scissor – No Function (20 mins)

Explain and Show – Rock, Paper, Scissor

- Next, we explain to students that for the next step, we will try to create a different form of game that uses randomized choices.
- In this case, we will create a rock, paper, scissors game. In this game, the player will face an opponent that will choose their moves randomly.
- We show the students how the game plays out and explain the game process such as being able to input 0, 1, or 2 as our answers and how the game will inform us if we win, lose, or must try again.
- We continue to analyze the project with the student and point out some important aspects that they might miss.

Discuss and Code – Rock, Paper, Scissor

- Step by step, we then explain our project to the students and ask them to add components that might be missing / if they are still confused, we can try to explain it slowly as the students try to follow.
- We discuss the project together from the very beginning and explain the code that we used to create the project. We explain that there are several key components in the code that they should know:

File/Code:

[4.RockPaperScissor\(NoFunction\).py](#)

- First, in the beginning we declared the necessary variables.
- Second, in the code, we created a function which will compare between the opponent and the player's answer.
- Third, Depending on the results, the program will then output different statements.
- For students who might be struggling, we can ask them to try and follow the code while we explain to them carefully about how it works.

Rock Paper Scissor – With Function (15 mins)

Discuss and Code – Rock, Paper, Scissor with Function

- Ask the students about the idea of making the program more efficient and readable.
- Ask the students if they're still remember about function in python and discuss if it's possible to implement function concept to the existing rock, paper, scissor project.
- Code together with the students to make the better version of the previous project by using function.
- Emphasize to the students that what they are doing is called refactoring, the process of making our code more efficient.

File/Code:

[5.RockPaperScissor.py](#)

Wrap Up (5 mins)

Wrapping Up

- We sum up by asking questions, making sure the students remember the most important things.
- We thank students for doing a great job overall.

☐ ☆ Good Luck With The Class !! ☆ ☐